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Availability of Alternative Prey Influences Avian Predation on Salmonids

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Abstract

The diet of two piscivorous bird species nesting in the lower Columbia River was compared with purse seine samples of the fish community to explore the influence of alternative prey availability on the consumption of juvenile salmonids in the Columbia River Estuary. The composition of Caspian tern (CATE) and double-crested cormorant (DCCO) diets broadly overlapped with each other and with independent purse seine sampling. However, consistent seasonal differences were observed in the mass of salmonid and marine forage fish prey in net sampling and the diets of piscivorous birds; Caspian terns had a higher proportion of salmonids in their diets than did cormorants or net sampling across seasons in all years. Caspian terns responded nonlinearly to the availability of marine forage fish, with the proportion of juvenile salmonids in their diet significantly declining when the proportion of marine prey species available exceeded a threshold of 48%. In contrast, cormorants showed a weak, linear response in this relationship. These results demonstrate how temporal variation in one prey type (marine forage fish that enter the estuary later in summer) can indirectly reduce consumption of another prey type (juvenile salmonids) by a common predator (piscivorous birds)—an apparent mutualism. This work shows the importance of understanding the broader food web. Developing management strategies that account for avian predator response to variation in the prey fish community will be important for increasing the survival of threatened and endangered salmonids in a changing environment.

Keywords Columbia River Estuary · Caspian tern · Double-crested cormorant · Salmonids · Forage fish · Prey availability

Introduction

In complex aquatic systems, predator–prey relationships often depend on how predators respond to the abundance and species composition of prey communities (Waraniak et al. 2017). In some cases, pulses of one species and a shared predator can result in lower mortality of another prey species (apparent mutualism; Webster and Almany 2002; Chaguaceda et al. 2020). In freshwater reservoirs, consumption of juvenile salmonids by fish predators varies considerably when alternate fish prey is available (Poe et al. 1991; Vigg et al. 1991). Off the central California coast, juvenile salmonids are subject to greater predation by common murres (*Uria aalge*) when juvenile rockfish abundance is low (Wells et al. 2017). Predation risk for juvenile salmonids from common murres and sooty shearwaters (*Ardenna grisea*) off the Washington and Oregon coast is markedly reduced when alternative prey such as northern anchovy are present (Phillips et al. 2021). While our knowledge of these relationships is often limited due to the lack of data on prey abundance and community composition, understanding these dynamics is essential for ecosystem-based management (Ainley et al. 2015) and species and ecosystem recovery (Chasco et al. 2017; Samhouri et al. 2017).

The effects of variation in prey assemblages on predator–prey interactions in the Columbia River Estuary are of ecological and natural resource management interest because it is thought that estuary avian predation on juvenile fish is one factor limiting the recovery of threatened and endangered Pacific salmon

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(*Oncorhynchus* spp.) populations. Every year during their seaward migration, over 100 million hatchery and wild juvenile salmonids from 13 listed populations migrate through the Columbia River Estuary to the Pacific Ocean (Weitkamp et al. 2012, 2015; Teel et al. 2014). During this migration, juvenile fish must travel past large breeding colonies of piscivorous Caspian terns (*Hydroprogne caspia*) and double-crested cormorants (*Nannopterum auritum*) that forage throughout the estuary (Collis et al. 2002; Lyons et al. 2007; Lyons 2010). To reduce avian predation and increase the survival of juvenile salmonids, there has been significant investment in research, monitoring, and management actions at predator colonies throughout the basin (USFWS 2005; USACE 2015; Roby et al. 2020). For example, the primary Caspian tern colony in the estuary was relocated from Rice Island (river km 34) to East Sand Island (river km 8), based on the reduced prevalence of salmonids and increased prevalence of marine forage fish in their diet on East Sand Island relative to on Rice Island (Collis et al. 2002; Roby et al. 2002).

While it is assumed that piscivorous birds in the Columbia River Estuary consume fish in accordance with their availability (e.g., Collis et al. 2002), an analysis of how avian predator diets change in response to changes in estuary prey availability of salmonid vs. alternative non-salmonid prey has not yet been conducted. The goal of this study is to explore the effect of variation in prey fish availability, measured via purse seine sampling, on predator diet composition from 2007 to 2012 during the period of juvenile salmon seaward migration. Specifically, we sought to (1) describe and compare the prey species assemblage sampled in the estuary with prey composition of Caspian tern and double-crested cormorant diets by season and year, (2) assess whether salmonid consumption changed relative to non-salmonid prey availability for either Caspian terns or double-crested cormorants, and (3) determine whether salmonid consumption response to marine forage fish availability was linear or nonlinear in nature.

Methods

Study Area

We conducted our study on fish prey availability and diet of two avian piscivores in the lower Columbia River estuary. This large estuary is located on the west coast of North America and runs into the Pacific Ocean between the states of Washington and Oregon. We conducted fish sampling at two stations (North Channel: 46° 14.2′N, 123° 54.2′W; Trestle Bay: 46° 12.9′N, 123° 57.7′W) in the estuary. We collected diet data for Caspian terns and double-crested cormorants during their nesting periods (April–July) from breeding colonies on East Sand Island, located approximately 5.5–6.0 km from estuary sampling stations (Fig. 1).

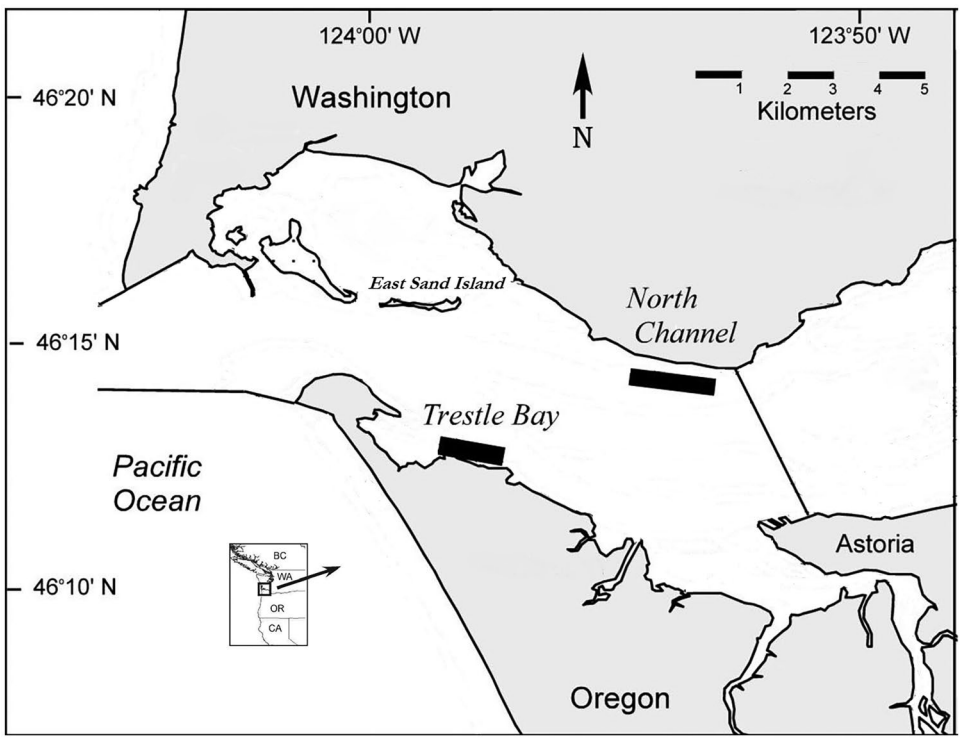


Fig. 1 Map of the lower Columbia River Estuary showing locations of purse seine sampling (Trestle Bay and North Channel) and bird diet sampling (East Sand Island). Inset map shows the location of Columbia River Estuary along West Coast of North America

Prey Availability Sampling

We compiled estuary fish assemblage data from a multi-year study designed to characterize the timing of salmonid outmigration of Columbia River basin stocks as well as the presence of other fish taxa in the Columbia River Estuary. The methodology is summarized here and described in detail in Weitkamp et al. (2012). While net sampling has biases, including differences in species-specific catchability (McCabe et al. 1983; Bottom and Jones 1990), it is still a useful method to characterize fish communities. Purse seine samples have been used to reflect general overall patterns of pelagic fish assemblages in the estuary (Weitkamp et al. 2012, 2015, 2016).

We conducted fish sampling at both estuary stations over 2 to 3-day-long trips ("cruises") every 2 weeks from mid-April to late June or early July during the years 2007–2012. We sampled fish during daylight hours using a fine mesh purse seine (10.6 m deep × 155 m long; stretched mesh opening, 1.7 cm; knotless bunt mesh, 1.5 cm) deployed in water approximately 10 m deep. Therefore, the net sampled roughly the entire water column, excluding benthic habitat. We repeatedly set and retrieved the net and processed catch throughout the incoming tide at each station, deploying the net in two configurations: (1) quantitative round hauls were set in a circle (area = 1912 m^2) and immediately pursed, and (2) nonquantitative tows that traveled upstream for 10 min before pursing the net (to increase the catch of fish). We estimated mean fish densities only from quantitative round hauls; to maximize sample sizes of fish lengths, we combined both quantitative round hauls and nonquantitative tows. We combined data from all quantitative round hauls from both sampling locations to represent each 2 to 3-day cruise, due to high across-locations and within-cruise catch variances (Weitkamp et al. 2012).

We identified all captured fish to species, counted them, and measured up to 50 individuals of each species (fork length [FL] or total length [TL] to the nearest 1 mm). Captured fish were, with the exception of sticklebacks, almost exclusively juveniles; any adult salmonids caught in the seine were excluded from analyses. For a subset of each fish species, we treated them with a lethal dose of MS-222; once onshore, we measured their length (FL or TL) and mass (grams). Steelheads were distinguishable from salmon species in purse seine samples and Caspian tern diet samples but not in double-crested cormorant diet samples; thus, some analyses of both avian piscivores combined steelhead and salmon species into "salmonids."

We categorized prey species into the following groups: salmon (Chinook (*Oncorhynchus tshawytscha*), coho (*O. kisutch*), chum (*O. keta*), sockeye (*O. nerka*)), steelhead (*O. mykiss*), northern anchovy (*Engraulis mordax*), clupeids (Pacific herring (*Clupea pallasii*), American shad (*Alosa sapidissima*), and Pacific sardine (*Sardinops sagax caerulea*)), surfperches (shiner perch (*Cymatogaster aggregate*), pile perch (*Rhacochilus vacca*), and striped perch (*Embiotoca lateralis*)), smelts (surf smelt (*Hypomesus pretiosus*) and longfin smelt (*Spirinchus thaleichthys*)), Pacific sand lance (*Ammodytes personatus*), flatfishes [Pleuronectidae], lampreys (Western river lamprey (*Lampetra ayresii*) and Pacific lamprey (*Entosphenus tridentatus*)), sculpins (Pacific staghorn sculpin (*Leptocottus armatus*) and prickly sculpin (*Cottus asper*)), sticklebacks (threespine stickleback (*Gasterosteus aculeatus*), and "other" (Table 1). We converted fish numbers to fish mass (fresh mass) using length-mass relationships derived from purse seine samples and whole fish collected on the Caspian tern colony (or published relationships for a few, rarely caught taxa). Prey groups with multiple species (e.g., clupeids included American shad, Pacific herring, and Pacific sardine) were the sum of masses of the individual species. We calculated the composition of estuary purse seine samples (total and percent mass for each prey category) for each cruise.

Predator Diet Sampling

We quantified the diet composition of Caspian terns nesting in the Columbia River Estuary during mate and chick provisioning (mid-April to mid-July) during the years 2007–2012. From observation blinds set up at the periphery of the tern breeding colony on East Sand Island, we used binoculars and spotting scopes to identify thousands of individual fish in bill-loads brought back by adults during twice-daily observation periods (high tide and low tide) both of which overlapped in part with estuary purse seine sampling). We identified observed prey to the lowest taxon possible (species or species group) and estimated total length (fork-length for some species) by comparing observed fish to the length of the Caspian tern's bill (~ 8 cm; Antolos et al. 2005). Observed fish prey were almost exclusively juveniles. We converted prey numbers to mass for each species or species group using length-mass relationships as described above for estuary purse seine sampling. Caspian tern bill loads are representative of prey purposed to feed chicks (Collis et al. 2002).

We quantified the diet composition of double-crested cormorants nesting on East Sand Island using identifiable soft tissue contents found in the foregut (esophagus, proventriculus, and ventriculus/gizzard) of adult cormorants lethally collected as they returned from foraging to provision chicks at the breeding colony (see Collis et al. 2002 for details). We identified prey from hundreds of foregut samples to the lowest taxonomic level possible using external features or diagnostic bones (dentaries, otoliths, and other bones aiding in the discernment of prey type) following artificial digestion of soft tissue. Foreguts lacking soft tissue (but possibly

Table 1 Prey categories used for estuary purse seine, Caspian tern diet, and double-crested cormorant diet samples

Prey category	Species	Prey group
Salmonids	Chinook salmon (*Oncorhynchus tshawytscha*)	Salmonids
	Coho salmon (*Oncorhynchus kisutch*)	
	Sockeye salmon (*Oncorhynchus nerka*)	
	Chum salmon (*Oncorhynchus keta*)	
	Steelhead[a] (*Oncorhynchus mykiss*)	
Anchovies	Northern anchovy (*Engraulis mordax*)	Marine forage fish
Clupeids	Pacific herring (*Clupea pallasii*)	Marine forage fish
	American shad (*Alosa sapidissima*)	
	Pacific sardine (*Sardinops sagax caerulea*)	
Surfperches	Shiner perch (*Cymatogaster aggregate*)	Marine forage fish
	Pile perch (*Rhacochilus vacca*)	
	Striped perch (*Embiotoca lateralis*)	
Smelts	Surf smelt (*Hypomesus pretiosus*)	Marine forage fish
	Longfin smelt (*Spirinchus thaleichthys*)	
Sand lances	Pacific sand lance (*Ammodytes personatus*)	Marine forage fish
Flatfish	Flatfish spp. [Pleuronectidae]	Estuarine/freshwater
Lampreys	Western river lamprey (*Lampetra ayresii*)	Estuarine/freshwater
	Pacific lamprey (*Entosphenus tridentatus*)	
Sculpins	Pacific staghorn sculpin (*Leptocottus armatus*)	Estuarine/freshwater
	Prickly sculpin (*Cottus asper*)	
Sticklebacks	Threespine stickleback (*Gasterosteus aculeatus*)	Estuarine/freshwater
Cyprinids[a]	Northern pikeminnow (*Ptychocheilus oregonensis*)	Estuarine/freshwater
	Peamouth (*Mylocheilus caurinus*)	
	Sucker spp. [Catastomidae]	
Others	Pipefish spp. [Syngnathidae]	Estuarine/freshwater
	Greenling spp. [Hexagrammidae]	
	Lingcod (*Ophiodon elongates*)	
	Pacific tomcod [Gadidae]	
	Gunnel spp. [Pholidae]	
	Prickleback spp. [Stichaeidae]	

[a]For the Caspian tern diet, steelhead is a distinct category, and cyprinids are included with "others"

containing bones), unidentifiable soft tissue lacking diagnostic bones, and bones not encased in soft tissue (e.g., at the bottom of the proventriculus) were excluded from the analysis.

Relative biomass of contributing prey types was based on the partitioning of soft tissue; for soft tissue that could not be identified via external (soft) tissue characteristics (due to partial digestion, absence of identifying body part, etc.), encased bones were used to assign soft tissue to appropriate prey categories. This assumes soft tissue digestion rates are similar among prey types (see Carss et al. 1997 for a comprehensive review of methods to determine proportional biomass breakdown of fish-eating bird diets). We identified from 21 to 25 kg of prey soft tissue fresh mass each year, which represented > 90% of total prey soft tissue mass, and we converted fresh mass to percent mass for each identified species or species group.

We compared fish data from estuary purse-seine cruises to diet composition data from piscivorous bird colonies from overlapping dates from 2007 to 2012 (Supplementary Information, Table S1). We calculated the total and percent mass for each prey category for Caspian terns and double-crested cormorants by collating diet data for up to 4 days before and after the dates of each purse-seining cruise. This resulted in 37 cruise/diet sampling periods for which we had both fish and bird diet data. To explore the effect of temporal pulses of salmonids and marine forage fish into the estuary (declining and increasing after late May, respectively; Weitkamp et al. 2016), we binned purse seine cruise samples, Caspian tern diet, and cormorant diet mass estimates into spring (April–May) and summer (June–July) cruises.

Data Analysis

To describe and compare the fish prey assemblage in the estuary with the diet composition of Caspian terns and double-crested cormorants between seasons and across years, we used nonmetric multidimensional scaling (nMDS), analysis of similarities (ANOSIM), and Dirichlet regression. We constructed nMDS plots using PRIMER-e (Clarke and Gorley 2015), calculating pairwise Bray–Curtis dissimilarity coefficients among purse seine, Caspian tern, and double-crested

cormorant samples for each cruise/diet sampling period using the square-root transformed percent mass values of each prey category (see Table 1). Each of the paired cruise/diet sampling periods was categorized by sampling dates into spring (April–May) or summer (June–July) seasons and plotted using the matrix of dissimilarity coefficients. This ordination technique places each coefficient in the nMDS space such that data points farther apart are less similar than data points closer together. This iterative algorithm is not guaranteed to converge to an optimal solution, so a number of restarts are run. Starting locations were random for each of the 25 iterations to find the best solution, as multiple runs of more iterations did not yield lower stress values; multiple iterations attained minimum stress, suggesting a true minimum solution. Stress values of <0.20 indicate that a spatial representation of data by an nMDS plot is consistent with the structure of the original data set (Clarke et al. 2014).

We evaluated the patterns observed in the nMDS plots using an analysis of similarities (ANOSIM in Primer-E), specifically testing the influence of sampling type, season, and year on the proportions of each prey category. This analysis produces global *R*-values that indicate the degree of separation of groups generated by a particular factor (or pair of factors). These global *R*-values range from 0 (no separation) to 1 (complete separation); the program also generates statistical probabilities by permutation.

To test whether proportions of broader functional and ecological prey groups varied across sampling types and time, we compared sample types (purse seines, Caspian tern diet, and double-crested cormorant diet) between seasons and across years using Dirichlet regression. We assigned the 13 prey categories into three main groups: salmonids included salmon and steelhead, marine forage species included schooling fishes generally found closer to the river mouth and ocean zone (anchovy, clupeids, surfperches, smelts, and sand lance), and estuarine species included largely non-schooling fishes distributed throughout the estuary mixing zone (lampreys, flatfishes, cyprinids, and others; Bottom and Jones 1990). We recalculated proportional mass for the three groups and tested whether the observed proportions of each group varied across sampling type, season, or year using the DirichletReg package (Maier 2021) in R (R Core Team 2020). The Dirichlet distribution represents a multivariate generalization of the beta distribution and allows for the simultaneous analysis of compositional data where the components sum to 1 (e.g., proportions of prey in diet) and the effects of covariates on this relative contribution of each component, without the need to transform the raw data (Douma and Weedon 2019). We used AIC values to evaluate a candidate set of models that varied in complexity using the observed proportions as the dependent variables (a matrix of the proportions of the three prey categories for each sampling type for each sampling period) and a combination of sampling type, season, year, and each of the two-way interactions as fixed effects. We considered models within two units of the lowest AIC value to have the strongest support and selected the model within this subset with the fewest explanatory parameters as the most parsimonious model (Burnham and Anderson 2002).

Our final set of analyses used a model selection approach to test whether (i) salmonid consumption for each avian predator was related to marine forage fish availability, (ii) any identified relationship was linear or nonlinear in nature, and (iii) a threshold value in marine forage fish availability could be identified. Nonlinear or threshold responses can identify where large changes in the response variable occur across small changes in the driver as well as potentially important reference points for management (Selkoe et al. 2015; Hunsicker et al. 2016). We compared linear and nonlinear models with and without temporal autocorrelation (Sonderegger et al. 2009; Bestelmeyer et al. 2011; Samhouri et al. 2012) using the proportion of salmon in the observed diet of each bird species as the response variable and the proportion of marine prey species available in the estuary as the independent variable for each sampling period across all years. Because our analyses focused on time series data, we first used a log-likelihood ratio test adapted from methods described in Gilman et al. (2012) and used in Samhouri et al. (2017) to determine whether a generalized additive mixed model with autocorrelated error structure (GAMM) was more appropriate than a generalized additive model (GAM) with normal error structure. Where temporal autocorrelation was determined to be significant, we used AICc to select between a GAMM ("gamm" function in R version 3.5.1 (R Core Team 2020)) and a linear model with autocorrelated structure (LMAC) in the residual covariance matrices. Where temporal autocorrelation was deemed non-significant, we used AIC*c* to select between a simple GAM (gam() function in the mgcv() package in R) and a linear model (LM). Detailed equations and structure for each of the four models are described in detail in Samhouri et al. (2017).

We determined whether there was more evidence for a linear or nonlinear model based on the following three criteria: (1) The estimated degrees of freedom, which increases with a curve's nonlinearity, was ≥ 1.8 (Zuur et al. 2009; Hunsicker et al. 2016); (2) the generalized cross-validation score (GCV; Wood (2004)) was minimized in the GAM compared to the LM (GCV is not a calculated criterion for GAMMs); and (3) the difference in AICc values was ≥ 2.0 units in favor of the nonlinear model (Burnham and Anderson 2002). For relationships identified as nonlinear, we defined the location of the threshold as the inflection point, that is, the value of the marine prey availability where the second derivative changed sign (Fewster et al. 2000; Bestelmeyer et al. 2011; Samhouri et al. 2012; Large et al. 2013). For these analyses, we calculated the 95% CI of the GAM smoothing function itself, along with its first and second derivative, via bootstrapping

of the residuals in order to allow for autocorrelation. This procedure generated a range of prey availability values where a threshold might occur. Because the location of the inflection point on the second derivative function is based on a statistical fit and cannot be determined exactly, we defined that location as the place where the second derivative is most different from zero within the determined range where the second derivative changed sign. Thus, we are most confident that the second derivative changed from zero to nonzero at this point.

Results

Predator Diet and the Estuary Fish Assemblage

During the spring and summer of 2007–2012, we conducted 37 estuary sampling cruises that overlapped in time with the sampling of diets of breeding Caspian terns and double-crested cormorants. We conducted 317 purse seine hauls and collected 340,995 fish during these cruises. We observed 16,333 fish prey of Caspian terns at the colony and collected and analyzed the contents of 730 double-crested cormorant foreguts (Supplementary Information, Table A1).

There was a broad overlap in the prey fish taxa observed among the three sampling types, particularly when compared within seasons; however, the proportions of each prey category did vary among the sample types (Fig. 2). The majority of fish observed were salmonids (salmon and steelhead) and marine forage fish (anchovy, clupeids, smelt, surfperch, and sand lance), which accounted for approximately 95%, 85%, and 60% of samples in the Caspian tern diet, estuary purse seine, and double-crested cormorant diet, respectively. The double-crested cormorant diet contained some fish taxa rarely observed in estuary purse seine sampling or Caspian tern diet, including flatfishes, cyprinids, and occasional taxa included in the "other" category (Table 1).

Multivariate analyses showed clear differences in the proportional composition of prey fish assemblages among sampling types and seasons. Using fine-scale prey categories, results in the nMDS plot showed distinct clustering by sampling types and further clustering of seasons within each sampling type (Fig. 3). Sampling type explained most of the variation (ANOSIM: Global $R = 0.70$, $p = 0.001$). Adding season marginally improved the model and led to better defined groups for sample type (Global $R = 0.76$) and season (Global $R = 0.37$, $p = 0.001$), while adding year did not improve the model (sample type: Global $R = 0.68$, $p = 0.001$; year Global $R = 0.02$, $p = 0.25$). While season was significant, the single variables of season and year did not explain much variation (season Global $R = 0.15$, $p = 0.001$; year Global $R = 0$, $p = 0.48$).

For the analyses using Dirichlet regression and three major prey groups, we found generally similar results to the prey categories analyses, with proportions of prey groups varying among sampling types and season consistently across years (Fig. 4). Two regression models provided equally good fits to the data, but the model with fewer parameters had more than twice the weight of evidence than the more complex model (65 to 25%; Table 2), so we used this model to describe the data. This model did not include "year," suggesting a consistent annual pattern of proportions of these prey groups across this period of years and despite some evidence that "year" and a "sampling type × year" interaction was important in the second-ranked model. The proportion of salmonids found in the diets of both species and in purse seine samples (sample types) was consistently greater in spring than in summer (Fig. 4; Table 3). Proportions of marine species were not consistent across all sampling types and seasons, as double-crested cormorants had relatively equal proportions in both seasons, while Caspian terns and purse seines had higher proportions of marine species in the summer (Fig. 4; Table 3). Proportions of estuarine species were very low in the diets of Caspian terns in both seasons, while double-crested cormorants preyed on these species at higher rates than observed across the other sampling types, particularly in the summer (Fig. 4; Table 3).

Relationship Between Salmonid Consumption and Marine Fish Availability

Our evaluation of responses of avian predators to prey availability found the GAM best explained the relationship between the proportion of salmonids in the Caspian tern diet and the proportion of marine prey species available in the estuary (Table 4), while a linear model with a negative slope best explained this relationship for double-crested cormorants (Table 4). The nonlinear response for Caspian terns (Fig. 5a) described how the proportion of salmonids in their diet significantly declined when the proportion of marine prey species in purse seine samples was greater than 0.48–0.65 (Fig. 5c), with the most likely location of this threshold at 0.48. In contrast, the significant ($p = 0.03$), but weak ($R^2 = 0.10$, deviance explained = 0.13) linear response for double-crested cormorants described how the proportion of salmonids in their diet declined relatively consistently as marine forage fish availability increased (Fig. 5d).

Discussion

These analyses mark the first examination of the diets of two piscivorous avian predators relative to the fish community in the lower Columbia River Estuary, despite both predators and fish prey in this river system being relatively

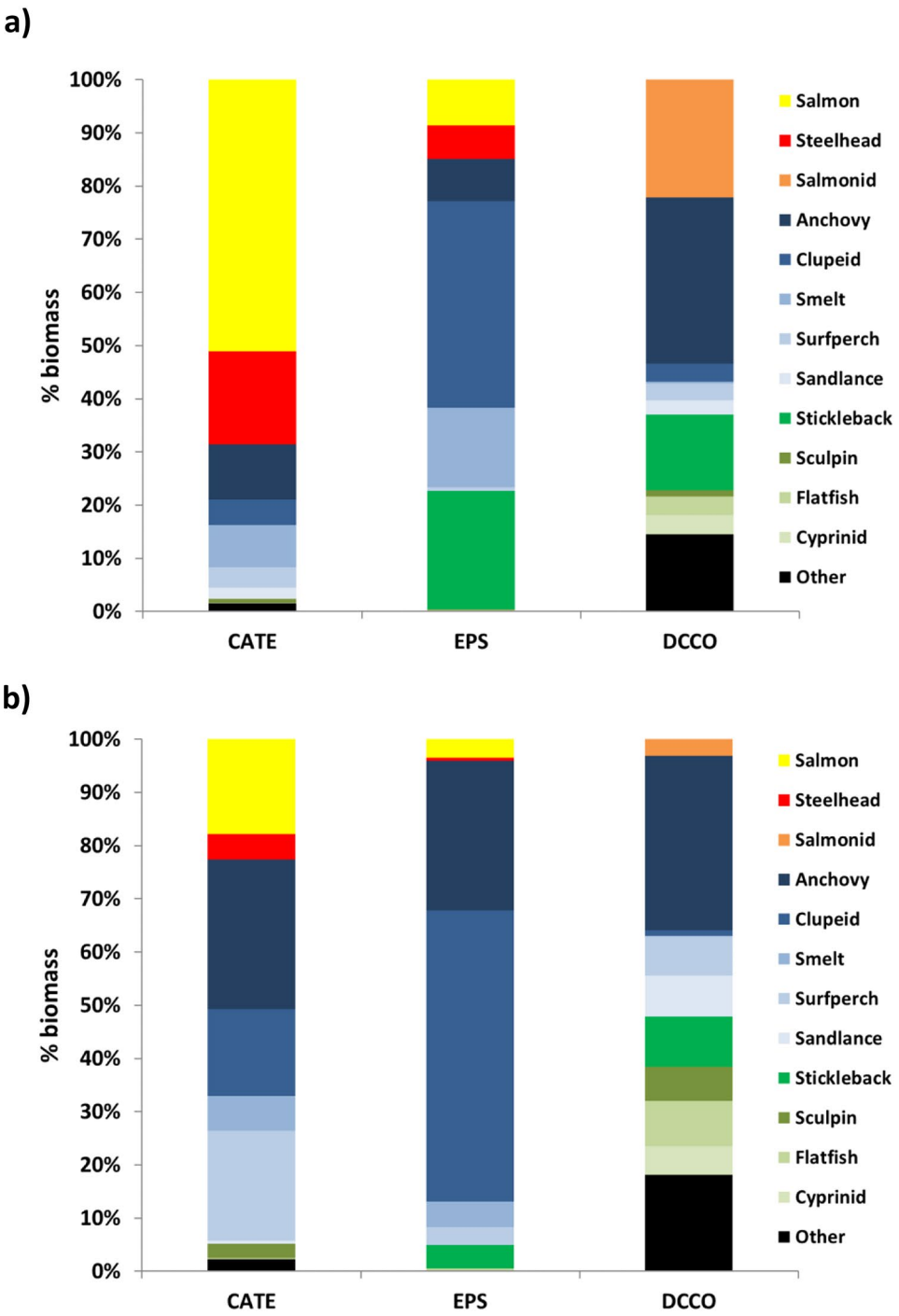


Fig. 2 Percent mass calculated for 13 prey categories from Caspian tern diet observations from the East Sand Island colony (CATE), purse seine sampling in the Columbia River Estuary (EPS), and double-crested cormorant foregut contents from the East Sand Island colony (DCCO) in **a** spring and **b** summer for combined years 2007 to 2012. Salmon (Chinook, coho, sockeye) and steelhead trout shown separately for CATE and EPS samples and combined into "salmonid" for DCCO samples

well-studied. The results highlighted differences in the seasonal availability of salmonids and other taxa within the fish community and identified relationships between predator diets and prey availability that suggest differential responses of the birds to salmonid and marine forage fish prey. These results suggest that contemporaneous monitoring of predator diet and available prey might provide important context when making conservation and management decisions (Lyons et al. 2014).

The broad overlap in prey and predator sampling supports the assumption that Caspian terns and double-crested cormorants are generally sampling the available estuary fish community, as reflected by purse seining. Both species spend time foraging in the marine or estuarine mixing zones of the estuary (Lyons et al. 2005, 2007; Anderson et al. 2004), where most of the fish prey is available and the purse seine sampling occurred. Schooling marine fishes found in both bird species' diets are available in shallow, nearshore

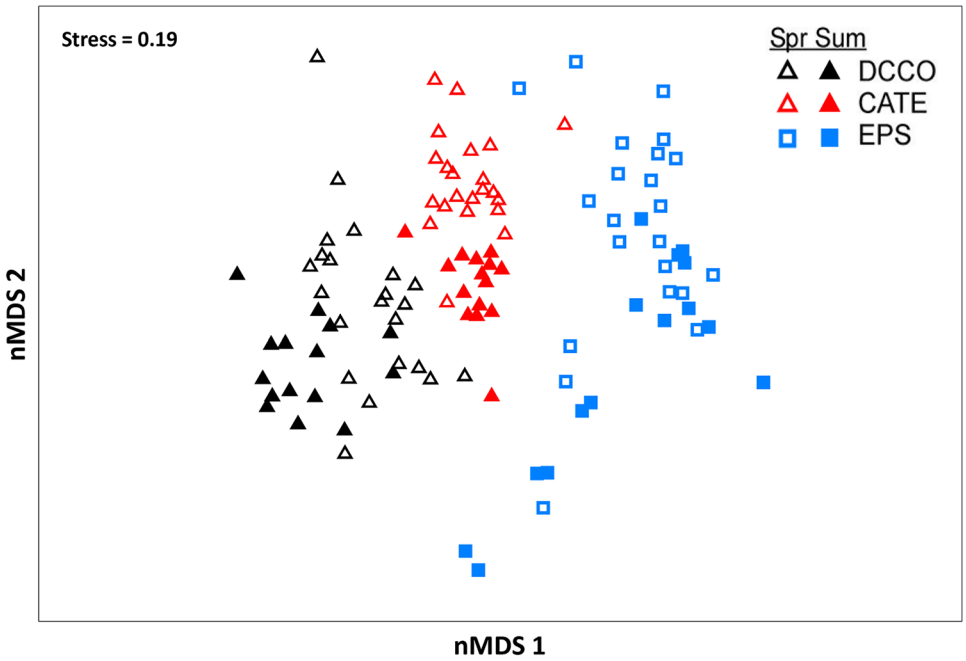


Fig. 3 Multidimensional scaling map of fish communities sampled via estuary purse seining (EPS; blue squares), the observed diet of Caspian terns (CATE; red triangles), and foregut samples of double-crested cormorants (DCCO; black triangles) during spring (open symbols) and summer (solid symbols) cruises from 2007 to 2012. The stress value (< 0.20) suggests the MDS representation is consistent with the original data structure and groupings shown were not made by chance (Clarke et al. 2014)

waters (Roegner et al. 2016) as well as in shallow waters of the Columbia River plume (Emmett et al. 2004). This latter area of freshwater and marine mixing can be up to 35,000 km^2 in size (Phillips et al. 2021) and is just over ~ 8 km from the colony. Broad overlap in seabird diet and prey assemblage observed in other U.S. west coast studies (Wells et al. 2017; Phillips et al. 2021) justify seabird diets as indicators of prey availability (Buren et al. 2012; Horn and Whitcombe 2015; Cavallo et al. 2020).

Our estimates of Caspian tern and double-crested cormorant diets confirmed differences between the species. While salmonids and marine forage taxa comprised most (85%) of the estuary purse seining overall, they comprised nearly all of the Caspian tern diet (97%) but less of the double-crested cormorant diet (59%), which included substantial estuarine/freshwater taxa (41%). Our estimates of the proportion of salmonids in the Caspian tern diet (~ 50%) exceeds previous estimates from this colony (~ 31% since 2001; Roby et al. 2013). One reason may be that we estimated percent mass rather than percent number, and salmonids, especially steelhead, are the largest and heaviest prey in our study. While diet based on mass and number are qualitatively similar, using mass better represents the relative importance of prey to the chick recipients of the prey deliveries. Another reason for the difference may be that previous estimates were based on sampling into late July when salmonids are less common in birds' diets (Roby et al. 2013).

Seasonal differences in prey availability seemed to influence diet patterns observed in both bird species. All three sampling types showed clustering by season in the nMDS analysis. This seasonal effect was reinforced by the regression analysis of season and sample types on the three major prey groups. A seasonal transition from salmonid- to marine taxa-dominated samples in channel habitats (this study) has also been observed for shoreline habitats in the estuary (Roegner et al. 2016), and it reflects lower numbers of yearling outmigrants of most salmonid stocks in the Columbia River system after early June (Weitkamp et al. 2015; Columbia River DART 2021).

This transition from salmonids to marine forage taxa noted for both bird species (Roby et al. 2002; Lyons et al. 2007) suggests tracking of prey resources but in different ways. For Caspian terns, the threshold of marine forage taxa above which salmonid consumption declines roughly coincides with a shift from the nest initiation and incubation period (April and May) to the chick-rearing and post-breeding period (June and July; Lyons et al. 2007). Diet transitions track the appearance or rapid growth phases of chicks in other terns and gulls (Annett and Pierotti 1989; Shealer 1998; Yakola 2019). In our case, while juvenile salmonids are a predictable resource for adults provisioning chicks, energetically profitable taxa such as large herring and anchovy (Schrimpf et al. 2012) are relatively more available in the summer.

Avian predator foraging behavior likely explains some of the observed diet patterns. Caspian tern foraging is confined to the top ~ 1 m of the water column (Cuthbert and Wires 2020); shallow, nearshore waters harbor schooling marine fishes (Emmett et al. 2004; Roegner et al. 2016) and salmonids, especially surface-oriented, hatchery steelhead (Johnson et al. 2000; Weitkamp 2008). Double-crested cormorant foraging reflects a generalist, pursuit-diving

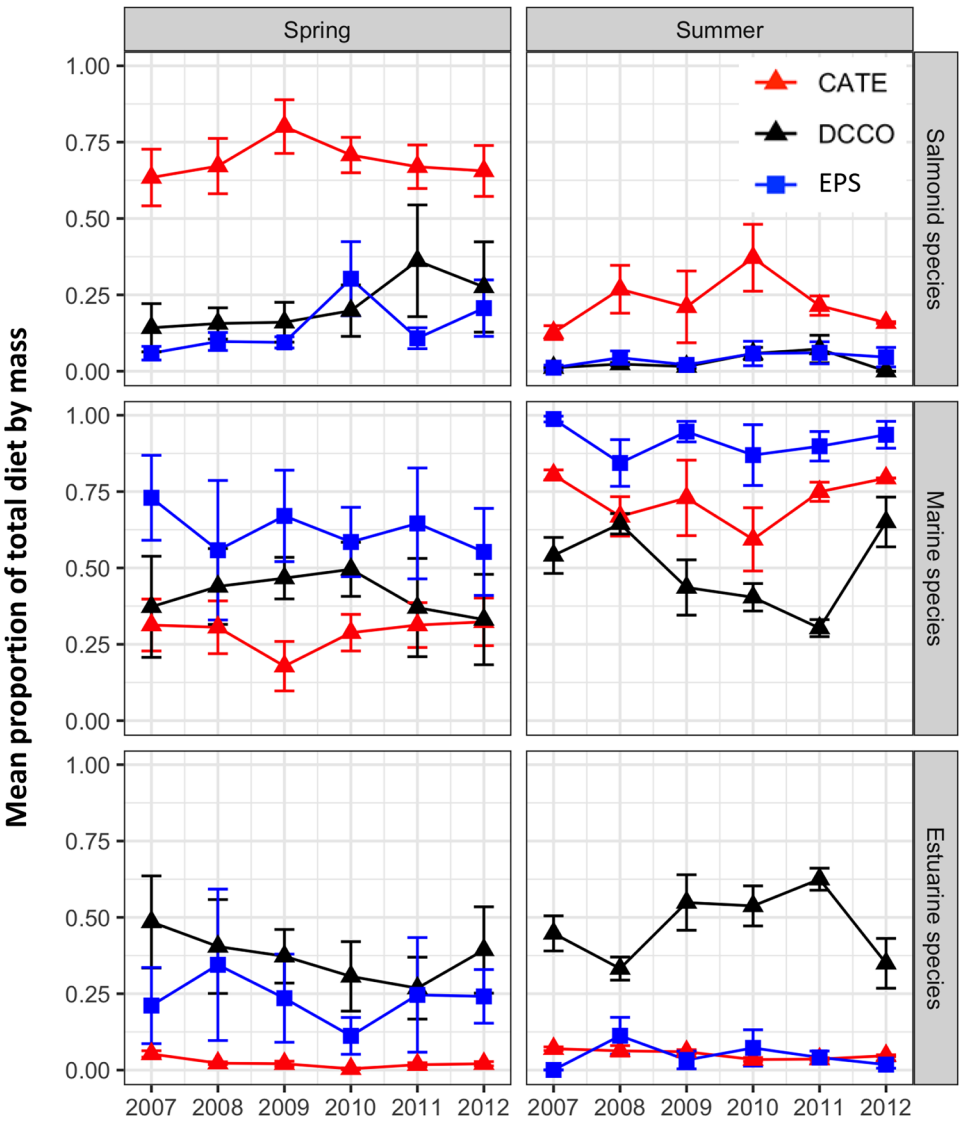


Fig. 4 Mean proportion (+/− se) of salmonid species (top), marine species (middle), and estuarine species (bottom) in the diet of Caspian terns (CATE), the diet of double-crested cormorant (DCCO), and estuary purse seining (EPS) during spring (April–May) and summer (June–July) cruises from 2007 to 2012

predator on pelagic and demersal prey (Dorr et al. 2020). Flatfish, sculpin, and other taxa are abundant in the beach (but not purse) seine sampling in the estuary (Roegner et al. unpubl. data), and sticklebacks are increasingly common due to severely modified river flow and reduced spring freshets from the dammed upper reaches of the Columbia River (Weitkamp et al. 2012). A diet that is 14% sticklebacks and nearly 25% flatfish, cyprinids, and "other" species (e.g., gunnels, suckers, pricklebacks, and lingcod) supports the idea that double-crested cormorants exploit

Table 2 The 95% confidence set of candidate models describing the relationship between prey fish groups (proportional biomass; $n = 111$) and sampling type (Caspian tern diet, double-crested cormorant diet, estuary purse seine), season (spring, summer), and year (2007–2012)

Model	log(L)	*K*	AIC	ΔAIC	ω_i
Sampling type + season + season × sampling type	259.0	18	−481.9	0.0	0.652
Sampling type + season + year + season × sampling type + sampling type × year	267.0	27	−480.0	1.9	0.252
Sampling type + season + year + season × sampling type	259.7	21	−477.5	4.4	0.072

A total of 18 Dirichlet regression models were evaluated, representing all possible combinations of three predictor variables and their two-way interactions. log(L) = log-likelihood, *K* = number of estimated parameters, AIC = Akaike's information criterion, ΔAIC = AIC for each model minus the lowest AIC from all possible models, and ω_i = model Akaike weight

Table 3 Statistical results of the most parsimonious Dirichlet regression model describing the relationship between proportions of three broad ecological fish prey groups and sampling type

Prey group	Factors	Estimate	*SE*	*Z*-value	*p*-value
Salmonids	(Intercept)	2.402	0.233	10.308	< 0.001
	Season (summer)	−1.010	0.356	−2.840	0.005
	Sampling type (DCCO)	−2.796	0.322	−8.679	< 0.001
	Sampling type (EPS)	−2.740	0.323	−8.484	< 0.001
	Season:sampling type (summer:DCCO)	1.029	0.494	2.082	0.037
	Season:sampling type (summer:EPS)	1.275	0.499	2.555	0.011
Marine	(Intercept)	1.520	0.230	6.595	< 0.001
	Season (summer)	1.095	0.356	3.077	0.002
	Sampling type (DCCO)	−1.484	0.325	−4.563	< 0.001
	Sampling type (EPS)	−0.772	0.334	−2.311	0.021
	Season:sampling type (summer:DCCO)	0.868	0.508	1.708	0.088
	Season:sampling type (summer:EPS)	0.945	0.522	1.810	0.070
Estuarine	(Intercept)	−0.360	0.216	−1.667	0.096
	Season (summer)	0.711	0.340	2.090	0.037
	Sampling type (DCCO)	0.513	0.317	1.618	0.106
	Sampling type (EPS)	−0.003	0.311	−0.008	0.993
	Season:sampling type (summer:DCCO)	1.124	0.498	2.256	0.024
	Season:sampling type (summer:EPS)	−0.561	0.487	−1.152	0.250

CATE, Caspian tern diet; *DCCO*, double-crested cormorant diet; *EPS*, estuary purse seine net sampling), season (spring, summer), and the interaction between sampling type and season

demersal resources based on prey availability (Gagliardi et al. 2015; Dorr et al. 2020).

Relative foraging locations of the two predators may also influence observed diet patterns. Caspian terns primarily forage near the colony in the marine or estuarine mixing zones and near the purse seine sampling sites, while double-crested cormorants spend more time in the freshwater zone upriver of the purse seine sampling sites (Lyons et al. 2005, 2007; Anderson et al. 2004). The latter situation is especially true in the spring, which likely only enhances double-crested cormorant reliance on salmonids. Still, Caspian terns that do forage upstream enhance their reliance on salmonids, as townet sampling at upstream sites finds few fishes other than salmonids in surface waters (Weitkamp et al. 2020). Double-crested cormorants may be obtaining demersal taxa upstream and marine taxa in the marine-mixing zone or in the Columbia River plume (Emmett et al. 2004; Litz et al. 2013). Thus, purse seine sampling appears to have more effectively gauged prey availability for Caspian terns than for double-crested cormorants.

Ever since the Caspian tern colony was relocated closer to the estuary mouth more than 20 years ago, there has been less reliance overall on salmonids and more reliance on alternative marine and estuarine fish taxa (Roby et al. 2002; Anderson et al. 2007; Lyons 2010). However, our analyses show that it takes relatively large pulses of marine forage fish (> 50% of purse seine mass) to observe a decline in salmonids in the Caspian tern diet. The timing of this change in

Table 4 Model selection results comparing linear and nonlinear models of the relationship between the proportion of salmon in CATE and DCCO diet and the proportion of marine prey sampled concurrently in purse seines. Bold models were selected as the best based on methodological criteria

	Model	AIC*c*	logLik	DE	edf	*p*	R^2	GCV	ΔGCV	ΔAIC*c*
CATE	GAMM	4.67	2.67	36	2.0	0.00	0.32	NA	NA	−2.00
	LMAC	6.67	1.67	22	35	0.00	0.20	NA	NA	NA
	GAM	**−1.40**	**5.17**	**36**	**1.9**	**0.00**	**0.32**	**0.05**	**−0.01**	**−4.42**
	Linear	3.02	1.67	22	35	0.00	0.20	0.06	NA	NA
DCCO	GAMM	−12.61	11.31	13	1.0	0.03	0.10	NA	NA	0.00
	LMAC	−12.61	11.31	13	35	0.03	0.10	NA	NA	NA
	GAM	−15.84	11.72	15	1.2	0.03	0.12	0.04	0.00	0.42
	Linear	**−16.26**	**11.31**	**13**	**35**	**0.03**	**0.10**	**0.04**	**NA**	**NA**

GAMM, generalized additive mixed model; *LMAC*, linear model with autocorrelated error structure; *GAM*, generalized additive model; *LM*, linear model; *AICc*, Akaike's information criterion corrected for small sample size; *logLik*, log likelihood; *DE*, deviance explained (%); *edf*, estimated degrees of freedom; *GCV*, generalized cross-validation score

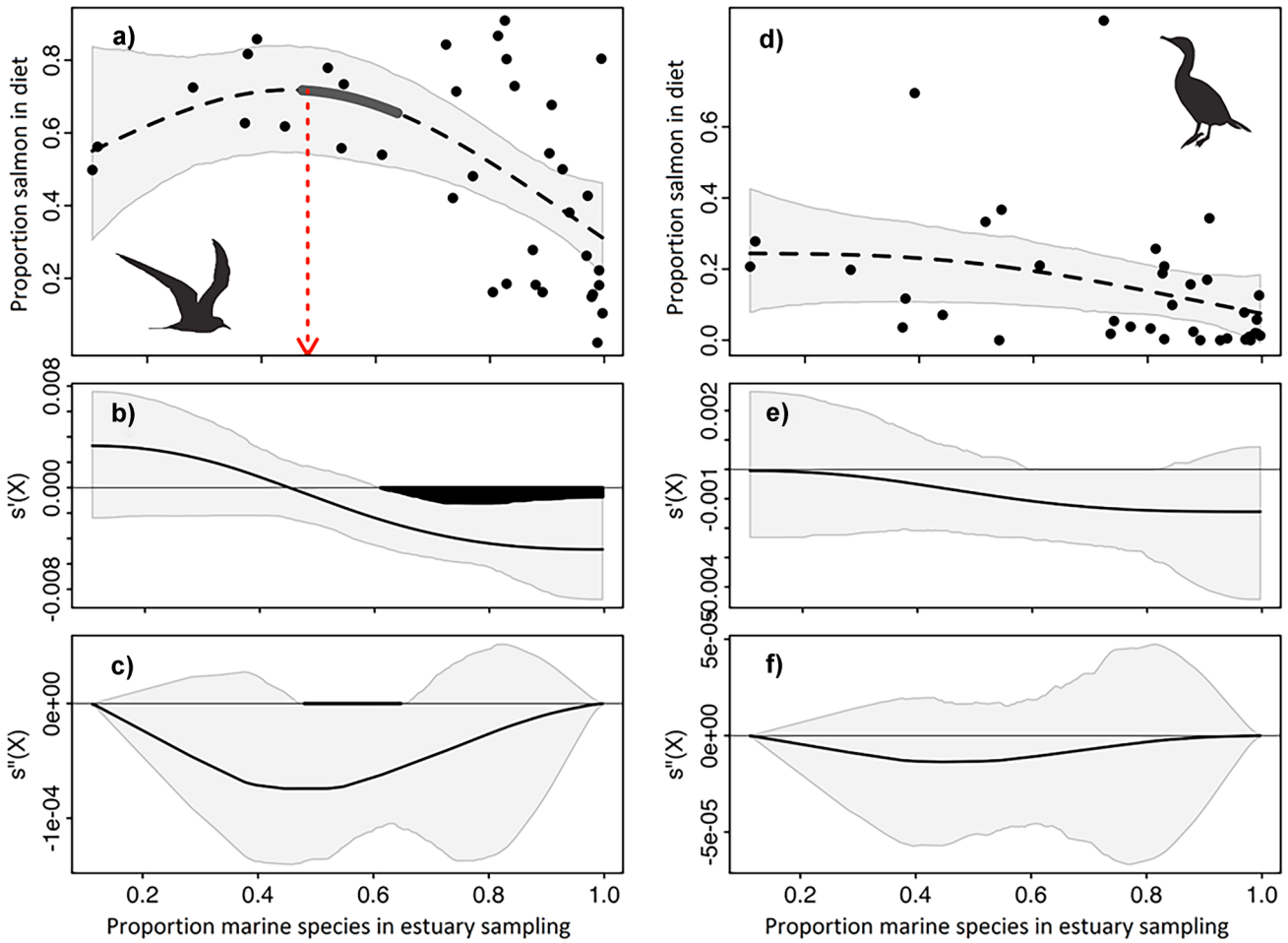


Fig. 5 Analyses of relationships between the proportion of salmonids in **a** Caspian tern and **d** double-crested cormorant diet, **b**, **e** 1st derivative s′ (X), and **c**, **f** 2nd derivative s″ (X) and proportion of marine prey species in concomitant purse seine samples for cruises in 2007–2012. The dashed line is GAM smoother, the gray polygon is 95% CI, points are cruise-level data, thick solid line is threshold range where 95% CI of the second derivative does not include 0, and red dotted arrow is the best estimate of the location of threshold (i.e., where the second derivative is at its absolute maximum value within the threshold range

relative prey availability in the estuary can vary over years but happens most often in June and July when northern anchovy and other marine forage fish move into the estuary.

For double-crested cormorants, a greater reliance on salmonids in spring relative to summer has also been reported (Roby et al. 2014). However, the spring reliance on salmonids is lower than on marine or estuarine fish in most years (Lyons et al. 2007), and cormorants often target salmonids when alternatives are relatively less common (Roby et al. 2013). Our analyses showed that while the cormorant diet appeared to track the seasonal availability of salmonids, it did not respond to pulses of marine forage fish (particularly in the summers of 2009–2011) and lacked a strong relationship or threshold. These patterns are consistent with a more generalist diet focused on a variety of marine and estuarine/freshwater fish. Similarly, in upriver Columbia River reservoirs, predation on juvenile salmonids by generalist fish predators including northern pikeminnow (*Ptychocheilus oregonensis*) and others varies with the availability of suckers [Catastomidae], sculpins [Cottidae], and shad (Poe et al. 1991; Vigg et al. 1991). Double-crested cormorants may also adopt a pelagic or demersal foraging strategy based on prey availability (Tollit et al. 1997). Such generalist tendencies suggest the food web may be buffered from the effects of predation by virtue of its diverse and abundant prey species and niches (Doucette et al. 2011). These tendencies suggest that double-crested cormorants might be an effective indicator species for the estuary fish community, while Caspian terns might be a good indicator for pelagic species.

Tracking of alternative prey is known for other marine birds. In the Gulf of California, Elegant terns (*Thalasseus elegans*) focus on Pacific sardine pre-recruits only after a threshold is reached; the relationship is such that the birds' use of sardine and anchovy can predict their catch in commercial fisheries later in the year (Velarde et al. 2015). In the Columbia River plume, the use of alternative prey such

as northern anchovy by piscivorous birds appears to mediate predation risk to juvenile salmonids (Phillips et al. 2021). In central California, when young-of-the-year rockfish are unavailable, seabirds turn to other forage species (Ainley et al. 1996; Webb and Harvey 2015). Northern anchovy prey, more coastal than rockfish in years of weak upwelling, are intermingled with out-migrating juvenile Chinook salmon, which leads to increased predation on the latter by piscivorous birds (Wells et al. 2017). Thus, northern anchovy acts as an apparent mutualist in one setting on the US West Coast and an apparent competitor in another. Pulses of productivity or alternative prey can produce such indirect relationships among prey in aquatic food webs (Webster and Almany 2002; Chaguaceda et al. 2020).

While this study did not analyze environmental factors, previous work in the estuary has demonstrated environmental effects on both the fish community and the avian predators. Variation in the Columbia River Estuary fish assemblage can be explained by river flow, plume volume, ocean temperature, and upwelling (Weitkamp et al. 2016). Northern anchovy, smelts, and clupeids make up half of the fishes sampled in normal river flow years but just a quarter of fishes sampled in years of high river discharge (Weitkamp et al. 2016). In fact, high river flow events can physically prevent the entry of marine forage fish into the estuary and reduce the residence time of both marine fish and salmonids (Weitkamp et al. 2012). Predation rates (% of available fish consumed) on tagged steelhead by Caspian terns are influenced by river flow, the North Pacific Gyre Oscillation, and regional upwelling, presumably through the mechanism of regulating the abundance and availability of marine forage fish in the estuary (Evans et al. 2016). In the high river discharge year of 2017, juvenile salmonids, anchovies, and smelts were replaced by lower-quality surfperches in the Caspian tern diet; many adults abandoned their nests and only returned to the colony to renest when anchovies became available in the estuary in July (Roby et al. 2018). Consumption of juvenile salmonids by double-crested cormorants is influenced by environmental factors as well as colony size; river discharge levels and the North Pacific Gyre Oscillation may influence the entry of marine prey into the estuary (Lyons et al. 2014). Taken together, these studies suggest that environmental factors may influence salmonid consumption by avian predators through influencing prey availability.

Implications for Pacific Salmon Conservation in a Changing Climate

One question on the minds of resource managers is how changes to the overall fish community in a warming Columbia River Estuary will influence bird predation on, and recovery of, Pacific salmon. As the estuary fish assemblage may be much different in the future (Weitkamp et al. 2016), comprehensive management strategies should consider how changes in alternative prey availability may influence predation on salmonids. Climate projections portend higher Columbia River flows in winter and early spring as well as reduced saltwater intrusion into the estuary (Weitkamp et al. 2012); these conditions may reduce marine forage fish availability in the estuary and lead to even higher reliance on salmonids by avian predators. Great cormorants (*Phalacrocorax carbo*) in Europe show declines in marine species in their diet during times of high river discharge (Dias et al. 2012). An associated risk for juvenile salmonids comes from increased estuary residence time, due to outmigrants not being physiologically ready to enter saltwater, which prolongs exposure to avian predators (Sebring et al. 2013; Lyons et al. 2014). Managing for these scenarios is difficult; there is the limited ability of the hydropower system to manage early spring flows, as they are usually caused by low elevation snowmelt below the mainstem dams (Sherwood et al. 1990).

Availability of alternative marine prey to avian predators in the estuary may also be affected by future oceanographic variability (e.g., El Niños, marine heat waves), which likely influence regional distribution and abundance of marine forage fish and their potential movement into the estuary. During the 1997–1998 El Niño, northern anchovy abundance was depressed in the Columbia River plume (Emmett et al. 2006). The absence of anchovy in nearshore habitats was reflected in avian piscivore diets, and Caspian tern and double-crested cormorant diets were dominated by salmonids and clupeids (Collis et al. 2002). Some climate projections include more Pacific sardine in the northern California Current (Smith et al. 2021), which could potentially expand the alternative prey base just outside the estuary. On the other hand, pulses of sardine may result in numerical responses in avian predators or colocation of sardine and salmonids, which could increase consumption by avian piscivores (Wells et al. 2017). Still, other climate forces may influence these predator–prey relationships. Ocean warming-driven hypoxia is predicted to reduce northern anchovy habitat in the California Current; the resultant migration to offshore, more northerly, or better-ventilated surface waters may affect the summer movement of anchovy into the Columbia River Estuary (Howard et al. 2020). The interplay of ecological, anthropogenic, and environmental factors are likely to be increasingly important in this complex conservation conundrum, and this study highlights how alternative prey availability and specific predator responses to the prey field may inform a suite of management strategies (e.g., habitat restoration, river flow regulation, predator control) that hope to improve Columbia River salmonid survival.

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